

Chapter 7: Deadlocks in Operating Systems

In a multiprogramming environment, several processes may compete for a finite number of resources. Deadlock occurs when multiple processes wait indefinitely for resources held by each other.

A deadlock situation can arise if and only if four conditions hold simultaneously:

1. Mutual Exclusion: At least one resource must be held in a non-shareable mode.
2. Hold and Wait: A process must be currently holding at least one resource and requesting additional resources.
3. No Preemption: Resources cannot be preempted from a process holding them.
4. Circular Wait: A closed chain of processes exists such that each process holds at least one resource needed by the next process in the chain.

Deadlock Handling Strategies

Operating systems typically handle deadlocks using three main methods:

1. Deadlock Prevention: Designing the system so that one of the four necessary conditions cannot occur.
2. Deadlock Avoidance: Banker's Algorithm is used to dynamically ensure the system never enters an unsafe state.
3. Deadlock Detection and Recovery: Allowing the system to enter a deadlock state, detecting it, and aborting processes to recover.